FRESHMEN ENGINEERING DEPARTMENT
Course Name:: INTRODUCTION TO PROGRAMMING

Branch: Common to All **ACY: 2023-24**
Class: I B.Tech - I Semester **Course Coordinator: Mr. P.K.V Lal**

UNIT-I

10M Questions

1) Explain about the Basic organization of a computer. 10M
2) What is a program? Explain types of programming languages. 10M
3) (a) What is an Algorithm? Explain characteristics of an algorithm. 5M
   (b) Write an algorithm to exchange the values of two variable using a third variable. 5M
4) (a) What is a Flow chart? Explain various symbols used in flow chart. 5M
   (b) Design a Flow chart to find whether a given number is even or odd. 5M
5) Explain the compilation and execution process. 10M
6) Explain the basic structure of a C program 10M
7) Explain Datatypes in C language 10M
8) What is a Constant in C? Explain different types of C constants with examples. 10M
9) (a) What is a Variable? Explain the rules for naming the C variables. 5M
   (b) Write a C program to add given two integers and display the result. 5M
10) Explain about the basic Input and Output functions in C. 10M
11) Explain different C Operators with examples. 10M
12) (a) Explain about Type Conversion and Type Casting. 5M
    (b) Explain Time and Space complexities of algorithms. 5M

2M Questions

1) What is a computer? List the basic components of a computer.
2) What is a Program and Programming?
3) What is Software?
4) What is an Algorithm?
5) Write any two advantages of writing algorithms.
6) What is Flowchart?
7) What is Pseudocode?
8) What is Compiler?
9) What is a Keyword? Write any 4 C keywords.
10) What is an Identifier? Write any two rules of naming C identifiers.
11) What is variable declaration?
12) What is variable initialization?
13) Differentiate formatted and unformatted I/O.
14) Explain getchar( ) and putchar( ) functions.
15) Explain ternary operator in C.
16) What is Operator precedence and Associativity?
17) What is Explicit Type Conversion?
18) Evaluate e = a + (b * c) / d if a = 20; b = 10; c = 15; d = 5.
19) List the steps involved in Problem Solving.
20) Differentiate top down and bottom-up approaches.